

Western Blot Integrity Atlas

Full reference-proteome molecular-weight audit report with embedded figures and complete tables

Defensive use only. This report helps evaluate whether a claimed western blot result is plausible and what controls are needed. It must not be used to plan, enable, or disguise blot substitution or fabrication.

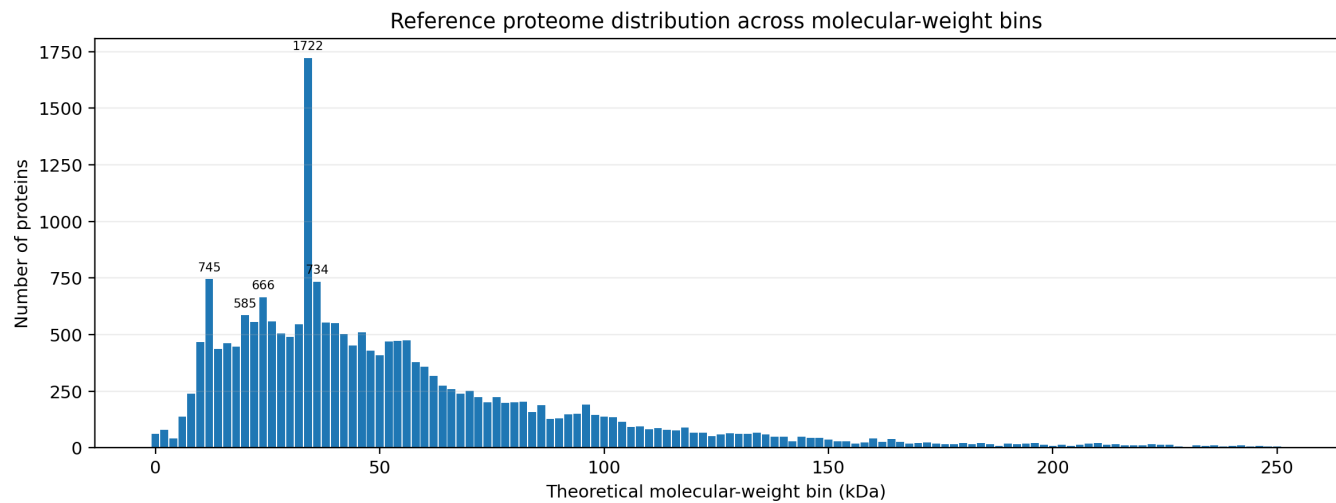
Metric	Value
Proteins parsed	21,853
Molecular-weight bins	266
Bin width	2 kDa
MW range	0.72 to 3906.49 kDa
Median MW	43.27 kDa
Mean MW	60.21 kDa
Proteins <= 50 kDa	12,475 (57.1%)
Proteins 50-100 kDa	6,394 (29.3%)
Proteins > 100 kDa	2,984 (13.7%)
High-priority bins	27
Elevated-priority bins	40
Routine-priority bins	199
Source FASTA	mouse_UP000000589_10090.fasta

Method in one breath

For each protein sequence, the script computes theoretical average molecular weight from residue masses, assigns a molecular-weight bin, applies the external marker, signaling-prefix, and family-pattern annotations, maps supplied public malpractice cases to bins, and calculates an audit-burden priority. Similar apparent molecular weight is never treated as proof of identity; it only increases the need for full-blot provenance and orthogonal controls.

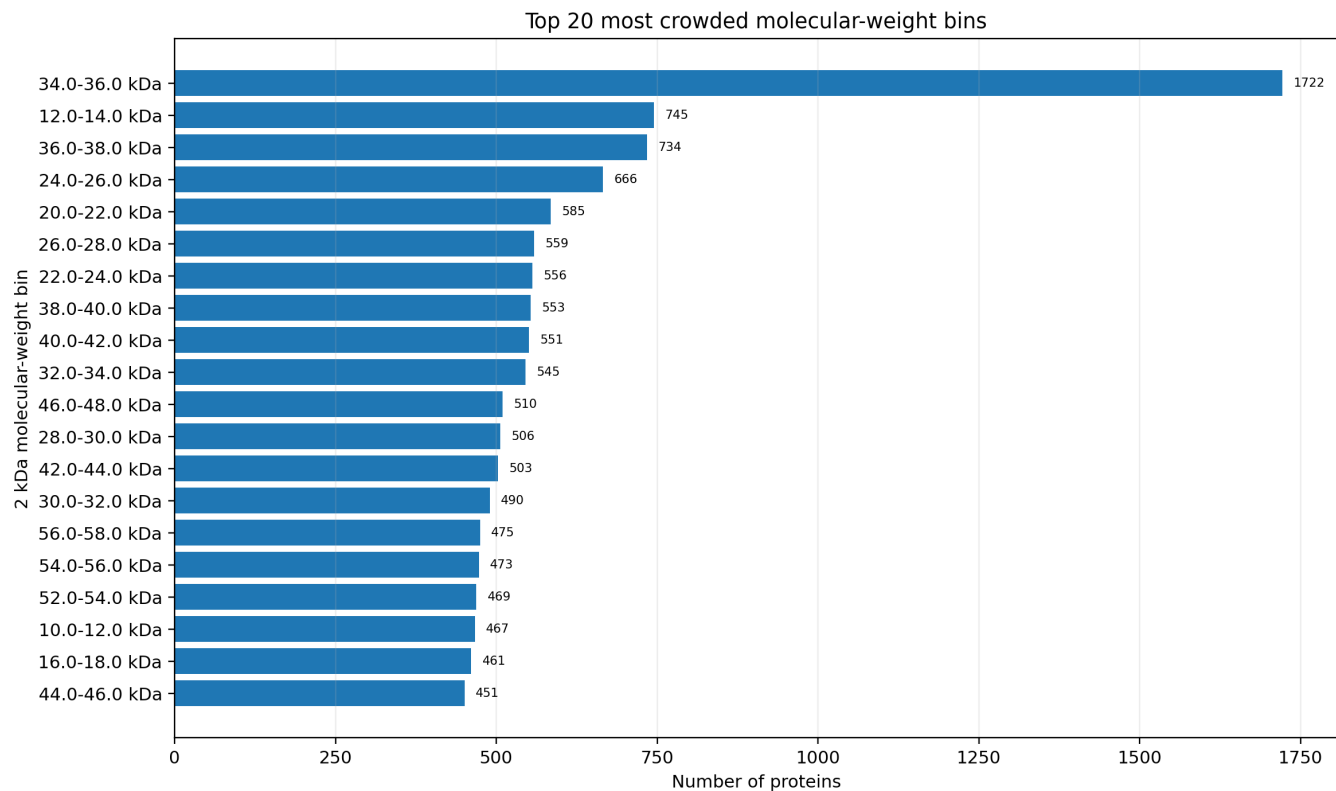
Molecular-weight distribution of the reference proteome.

Proteins were grouped into theoretical molecular-weight bins. Dense gel regions are especially relevant for western blot review because many unrelated proteins occupy nearby molecular-weight neighborhoods.



Most crowded molecular-weight neighborhoods.

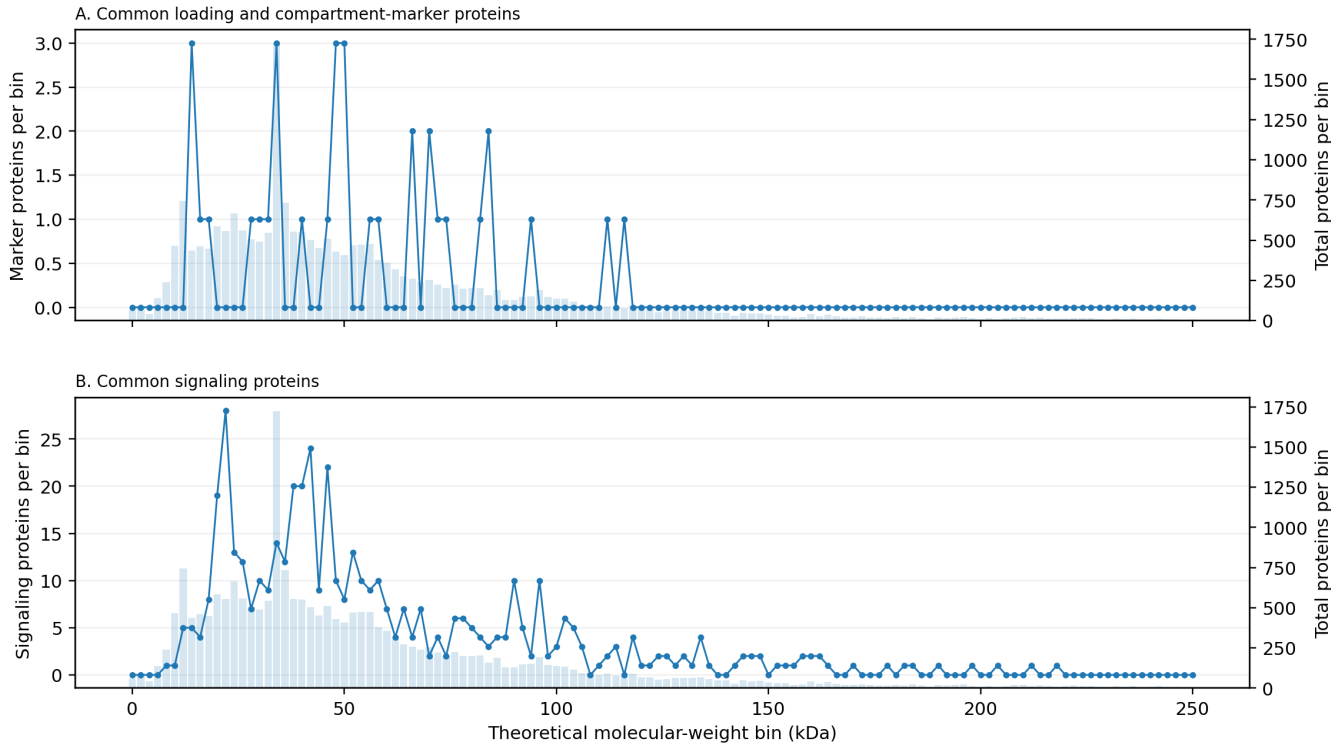
The top bins are ranked by protein count. These bins are not suspicious by themselves; they indicate where molecular weight alone is a weak identity criterion and where validation documentation becomes more important.



Distribution of common western blot markers and signaling proteins across molecular-weight bins.

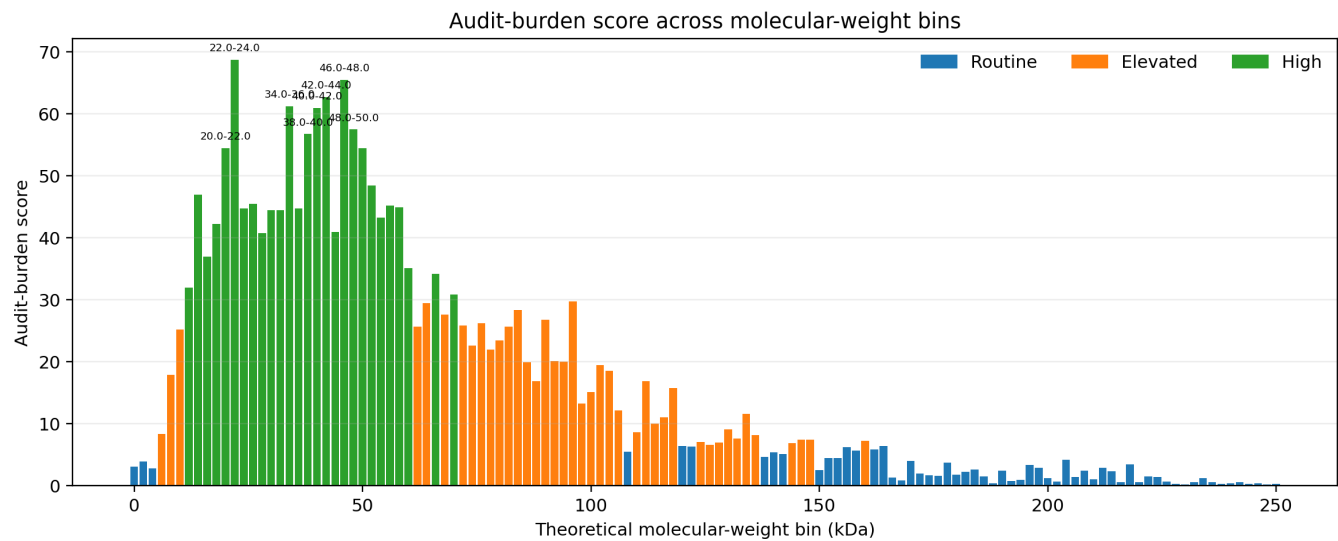
The pale background bars show total protein density, while the overlaid lines show audit-relevant annotations. Panel A shows configured loading or compartment markers; Panel B shows configured signaling-prefix matches.

Audit-relevant annotations across molecular-weight bins



Audit-burden score across molecular-weight bins.

The score integrates protein density, marker count, signaling-prefix count, and configured family/context information. High-scoring bins define high-documentation zones for western blot identity claims.



Highest-audit molecular-weight neighborhoods: score summary

MW bin	N	Markers	Signal	Priority	Score
22.0-24.0 kDa	556	0	28	high	68.8
46.0-48.0 kDa	510	1	22	high	65.5
42.0-44.0 kDa	503	0	24	high	62.8
34.0-36.0 kDa	1722	3	14	high	61.2
40.0-42.0 kDa	551	1	20	high	61.0
48.0-50.0 kDa	430	3	10	high	57.5
38.0-40.0 kDa	553	0	20	high	56.8
20.0-22.0 kDa	585	0	19	high	54.5
50.0-52.0 kDa	408	3	8	high	54.5
52.0-54.0 kDa	469	0	13	high	48.5
14.0-16.0 kDa	436	3	5	high	47.0
26.0-28.0 kDa	559	0	12	high	45.5
56.0-58.0 kDa	475	1	9	high	45.2
58.0-60.0 kDa	378	1	10	high	44.9
36.0-38.0 kDa	734	0	12	high	44.8
24.0-26.0 kDa	666	0	13	high	44.8
32.0-34.0 kDa	545	1	9	high	44.5
30.0-32.0 kDa	490	1	10	high	44.5
54.0-56.0 kDa	473	0	10	high	43.2
18.0-20.0 kDa	448	1	8	high	42.2
44.0-46.0 kDa	451	0	9	high	41.0
28.0-30.0 kDa	506	1	7	high	40.8
16.0-18.0 kDa	461	1	4	high	37.0
60.0-62.0 kDa	358	0	7	high	35.1
66.0-68.0 kDa	259	2	4	high	34.2

Highest-audit molecular-weight neighborhoods

These bins combine protein density with configured markers, signaling-prefix matches, and family-rich regions. They are not substitution recommendations; they are places where reviewers should be most insistent about source files and controls.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
22.0-24.0 kDa	556	0	28	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family	high	Ctf2, Ankub1, Rhoc, Lrrc51, Ctag2l2, Ap3s2, Gsc2, Ftdc2, Rcan2, Cbfb, Btf3, Ccdc28b, Commd3, Tceal5, Tmem125, Krtap10-10, Angp8b,
46.0-48.0 kDa	510	1	22	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	Sdc3, Serpina1a, Cdc71, Aadacl2fm2, Mppe1, Cers4, Abraxas1, Pdk2 [Mitochondrial markers], Sec14l4, Igsf11, Acatn1, Rtn4rl2, Cers3, Sec14l3, Nr2f1, Azi2, Rpl3, AAdac14fm3
42.0-44.0 kDa	503	0	24	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Actb12 [Actin family], Acta2 [Actin family], Foxo3, Art3, Opn5, Actc1 [Actin family], Septin1, Syne4, Gna11, Rnf34, Yy2, Hsd3b3, Fut9, Mapk12 [MAPK cascade], Sstr4, Ldhal6b, Acta1 [Actin family], Ttrh2
34.0-36.0 kDa	1722	3	14	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	Or4b1, Cd276, Slc39a3, Spry1, Tomm40l [Mitochondrial markers], Dhrs1, Or4c127, Ogn, Pex26, Or1j10, Ceacam12, Syp, , Or6d1b, Cdk5r1 [Cell-cycle CDK/cyclin], Or3a4, Katnb1, Vmn1r63
40.0-42.0 kDa	551	1	20	Actin family; Cell-cycle CDK/cyclin; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	Ptgdrr, Sstr5, Sp6, Plek2, Nans, Cxcr1, Lgals9, Gpr27, Rpf1, Septin3, Spn, Gjd4, Lhx6, Ccr8, Rbm4, Tob1, Plm2, Lrrc69
48.0-50.0 kDa	430	3	10	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Histone family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family	high	Dph1, Chst9, Rnf150, Kcnj16, Entr1, Krt23, Dok3, Tead4, B020011L13Rik, Tbccl [Tubulin family], Tbx19, Bzw1, Foxd4, Ccnb1 [Cell-cycle CDK/cyclin], Fam117a, Cmas, Pnma8a, Bzw2
38.0-40.0 kDa	553	0	20	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	Rcn3, Taar8b, Pax4, Borcs6, Cts6, Opclm, Rnf167, Ptgr2, Rnf39, Gmpr2, Wdr45b [Actin family], Acmsd, Scamp1, Tas2r123, Elmod1, Zfp696, Gimap4, Sl3gal4
20.0-22.0 kDa	585	0	19	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	Tmem240, Obp2b, Sat1, Urad, Ssr2, Jpt2, Park7, Itgb3bp, Csn3, Igf2, Faiml, Ssxh13, Reg3d, Faap20, Cenpm, Esm1, Smap, Bcl2a1d [Apoptosis caspase/BCL]
50.0-52.0 kDa	408	3	8	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	Serpina3f, Cdc91, Rundc3a, Adss2, Zfp113, LOC102638047, Cmpk2 [Mitochondrial markers], Cdc25c, Tuba8 [Tubulin family], Eef1g, Zfp36l2, Errf1, Hlx, Wipf1 [Actin family], Slc39a8, Slc46a1, Tubb6 [Tubulin family], Tgfb11
52.0-54.0 kDa	469	0	13	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Chia, Serpinc1, Fezf1, Cpsf7, Zfp874a, Gsdma3, Krtap16-1, Rmdn3, Tekt4, Wrap73, Atg4c, Snx8, Mettl17 [Mitochondrial markers], Glra2, Slc36a2, Nr5a1, Fam124b, Cables2
14.0-16.0 kDa	436	3	5	Actin family; Apoptosis caspase/BCL; Histone family; Lamin family; Mitochondrial markers; p53 pathway	high	Creb12, Trbv12-2, H2ac21 [Histone family], Txndc17, Lce1c, Gpha2, Lce1f, Lce1a2, Gabarapl1, H2aj [Histone family], Snob, H2ac1 [Histone family], Batf, Anp32c, Ly6g6c, , Trav15d-1-dv6d-1, Ly6c2
26.0-28.0 kDa	559	0	12	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	Mpig6b, Mdfi, Lat, Malsu1 [Mitochondrial markers], Clns1a, Tmed4, A1597479, Them4, Mob4, Rnd1, Tirap, Ceacam9, Gm20812, Prizc1 [Actin family], Zfyve21, Gm20834, Tspan4, Mpzf3
56.0-58.0 kDa	475	1	9	AKT pathway; Actin family; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Mitochondrial markers; Nuclear receptor PPAR	high	Usp14, Irf5, Klc3, Cyp2c70, Cyp2c66, LOC620551, Traf2, Slc18a1, Slc43a3, Rnf217, Cyp2t4, Gcm2, Nagpa, Zfp13, Matk, Cyp2c68, Slc1a4, Pramel45
58.0-60.0 kDa	378	1	10	Actin family; Cell-cycle CDK/cyclin; ER markers; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	Cct6a, Ccnl2 [Cell-cycle CDK/cyclin], Fto, Cyp19a1, Cyp3a59, Crnn, Agf1g, Gm10778, Lao1, Kihl33, Cct4, Znf513, Hinfp [Histone family], Sntb1, Zfp273, Slc29a4, Rorc, Vrk2
36.0-38.0 kDa	734	0	12	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	Vmn1r193, V1ra11, Ankrd1, Or2aj6, Or5al5, Frzb, Or9k7, Prss32, Vmn1r84, Tas2r122, Or4a27, Vmn1r227, Or14j6, Alad, Gm5565, Srx20, Ammcr1, Wdr5b
24.0-26.0 kDa	666	0	13	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Lamin family; Mitochondrial markers; p53 pathway	high	Zfand6, Nme5, Ropn1, Yae1d1, Aicda, Ndufs8 [Mitochondrial markers], Coq7 [Mitochondrial markers], Tpd52l2, Cbln2, Tex35, Csprs, Orm3, Mospd1, Muc3a, Evi2a, Cabccoco1, Cldn24, Ms4a4b
32.0-34.0 kDa	545	1	9	AKT pathway; Actin family; Cell-cycle CDK/cyclin; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	Zmat3, Gjb1, C1qtnf1, Oser1, Nkx1-2, Tlpa, D17h653e, Sh2b6, Btdb19, Dnase1, Mrgprg, Isoc1, Bcdin3d, Pblid1, Elovf3, Hey1, Ssr1, Eglf8
30.0-32.0 kDa	490	1	10	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Lamin family; Mitochondrial markers	high	Svs3b, Sna2, Gm9758, Rchy1, Film2, Mapre1, 2610028H24Rik, Alyreffm10, Fcgr3, Nsa2, Speer4e2, , Pip4p1, Plpbb, , Dhrs2, Dnaq9, Dcoun1d2
54.0-56.0 kDa	473	0	10	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	Zfp810, Tmem200a, Etv4, Nab1, Tmem161a, Pnlrp32, F10, Ltv1, Polr1e, Pnma3, Krt24, Scarb2, Fam114a2, Insm1, Pkmyt1, Klf11, Zfp872, Abtb1
18.0-20.0 kDa	448	1	8	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Mitochondrial markers; Tubulin family	high	Spaca5, Ube2t, Jchain, Il36a, Ncbp2, Czib, Slx1, Pttglip2, Pmp22, Wfdc2, Gm16405, Cyrr1, 6030498E09Rik, Spata19 [Mitochondrial markers], Eola1, 2410137M14Rik, Ucn3, Ten1
44.0-46.0 kDa	451	0	9	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers	high	Foxh1, Mtg2 [Mitochondrial markers], Amt [Mitochondrial markers], Lhx3, Fnta, Npy1r, Vsig1, Sirpb1a, Lhx9, Mapkapk2 [MAPK cascade], Gna13, Gpr137b, Lef1, Nipal4, Tex9, Mvd, Zfp707, B4gal3
28.0-30.0 kDa	506	1	7	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel	high	Krtap4-27, Crybb1, Hoxb9, Pdc, Rpl8, Hdhd3, Tmt1a3, Tmem33, Wbp2, Amy2a1, Ccdc59, Pip4p2, N4bp21, Neur3, Kik15, Tmt1a2, PcgF3, Tspan3
16.0-18.0 kDa	461	1	4	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; p53 pathway	high	MNCb-2990, Rpl32-ps, Bspb1, Trappc2l, Sval2 [Actin family], Pthrd1, Vpreb1b, Carhsp1, Ascl4, Slc31a2, 1700029F12Rik, Il9, Jpt1, Rps19, Cnih4, Csf2, Cst9, Cox5a [Mitochondrial markers]
60.0-62.0 kDa	358	0	7	Actin family; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	Fmo5, Ugt1a9, Slc22a6, Mon1b, Slc16a2, Nxp5f, N4bp3, Tmem184c, Ugt1a1, Flvcr2 [Lamin family], Bmpr1a, Vangl1, Hpse, Traf6, Adgrg3, Ugt1a5, Rbm46, Pank1
66.0-68.0 kDa	259	2	4	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Lamin family; Mitochondrial markers	high	Gm8113, Zfp345, Zbtb22, Dcaf8, Frs1, Gga2, Plk5, Lrfn3, Hrob, Rap1gds1, C8a, Slc13a1, Lrrtm3, Sl7, Lsr, Cdk5rap1 [Cell-cycle CDK/cyclin; Mitochondrial markers], Gm35315, Enc1
12.0-14.0 kDa	745	0	5	Actin family; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers; Tubulin family; p53 pathway	high	, , , , , Pex39, Slurp1, Emc6, Dnal4, , , , , , Lymr7, Tpc5os, , , Tmem265
70.0-72.0 kDa	253	2	2	Actin family; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers	high	Chml, Plk3, Eng, Mbd1, Zfp997, Ncbp3, Nop9, Slc6a4, Slc23a2, Znym5, Hspa1a [Heat shock proteins], Mak, Glce, Rufy2, Il18rap, Epm2aip1 [Actin family], Zfp949, Zap70
96.0-98.0 kDa	191	0	10	Actin family; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel	elevated	Hrnpu1l, Polk, Gm61617, Ilf3, Pdc6lp [Actin family], Znf629, Sycp2l, Map3k12 [MAPK cascade], Kansl3, Vmn2r22, Map10, Tlc7a, Vmn2r14, Vmn2r26, Tyro3, Tik, Axin1, Strip2 [Actin family]
64.0-66.0 kDa	274	0	7	Actin family; Cell-cycle CDK/cyclin; Histone family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	elevated	Arlh1, Nr5a2, Zfp870, Rufy4, A330040F15Rik, Slc15a3, Cdkl2 [Cell-cycle CDK/cyclin], Zfp960, Fzd2, Rbm47, Znf354c, Znf879, , Gm6710, Zfp955a, Parbp [Actin family], Slc26a11, Slc13a2
84.0-86.0 kDa	158	2	3	AKT pathway; Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; Mitochondrial markers; NF-kB/Rel	elevated	Zfp184, Lima1 [Actin family], Dtna, Fam83c, Syt5f, Sox5, Arhgap24, Cdh24, Pthf2, Prss12, Ppp1r3f, Adam11, Dckl1, Glis1, Pcsk1, Zmat1, Haspin, Kank3

Common loading/compartment markers present in this proteome file

MW kDa	Gene	Accession	Entry	Protein	MW bin
14.06	H2ac1	L7MTW0	L7MTW0_MOUSE	H2A clustered histone 1	14.0-16.0 kDa
15.39	H3c2	P84228	H32_MOUSE	Histone H3.2	14.0-16.0 kDa
15.40	H3c1	P68433	H31_MOUSE	Histone H3.1	14.0-16.0 kDa
16.28	Tomm20	Q9DCC8	TOM20_MOUSE	Mitochondrial import receptor subunit TOM20 homolog	16.0-18.0 kDa
19.53	Cox4i1	P19783	COX41_MOUSE	Cytochrome c oxidase subunit 4 isoform 1, mitochondrial	18.0-20.0 kDa
28.78	Pcna	P17918	PCNA_MOUSE	DNA sliding clamp PCNA	28.0-30.0 kDa
31.73	Vdac2	Q60930	VDAC2_MOUSE	Non-selective voltage-gated ion channel VDAC2	30.0-32.0 kDa
32.35	Vdac1	Q60932	VDAC1_MOUSE	Non-selective voltage-gated ion channel VDAC1	32.0-34.0 kDa
34.22	Rplp0	P14869	RLA0_MOUSE	Large ribosomal subunit protein uL10	34.0-36.0 kDa
34.71	Tbp	P29037	TBP_MOUSE	TATA-box-binding protein	34.0-36.0 kDa
35.81	Gapdh	P16858	G3P_MOUSE	Glyceraldehyde-3-phosphate dehydrogenase	34.0-36.0 kDa
41.74	Actb	P60710	ACTB_MOUSE	Actin, cytoplasmic 1	40.0-42.0 kDa
47.99	Calr	P14211	CALR_MOUSE	Calreticulin	46.0-48.0 kDa
49.91	Tubb2a	Q7TMM9	TBB2A_MOUSE	Tubulin beta-2A chain	48.0-50.0 kDa
49.91	Tuba1c	P68373	TBA1C_MOUSE	Tubulin alpha-1C chain	48.0-50.0 kDa
49.95	Tubb2b	Q9CWF2	TBB2B_MOUSE	Tubulin beta-2B chain	48.0-50.0 kDa
50.14	Tuba1a	P68369	TBA1A_MOUSE	Tubulin alpha-1A chain	50.0-52.0 kDa
50.15	Tuba1b	P05213	TBA1B_MOUSE	Tubulin alpha-1B chain	50.0-52.0 kDa
50.42	Tubb3	Q9ERD7	TBB3_MOUSE	Tubulin beta-3 chain	50.0-52.0 kDa
57.06	P4hb	P09103	PDIA1_MOUSE	Protein disulfide-isomerase	56.0-58.0 kDa
59.75	Atp5f1a	Q03265	ATPA_MOUSE	ATP synthase F(1) complex subunit alpha, mitochondrial	58.0-60.0 kDa
66.79	Lmnb1	P14733	LMNB1_MOUSE	Lamin-B1	66.0-68.0 kDa
67.28	Canx	P35564	CALX_MOUSE	Calnexin	66.0-68.0 kDa
70.08	Hspa1a	Q61696	HS71A_MOUSE	Heat shock 70 kDa protein 1A	70.0-72.0 kDa
70.87	Hspa8	P63017	HSP7C_MOUSE	Heat shock cognate 71 kDa protein	70.0-72.0 kDa
72.59	Sdha	Q8K2B3	SDHA_MOUSE	Succinate dehydrogenase [ubiquinone] flavoprotein subunit, mitochondrial	72.0-74.0 kDa
74.24	Lmna	P48678	LMNA_MOUSE	Prelamin-A/C	74.0-76.0 kDa
83.28	Hsp90ab1	P11499	HS90B_MOUSE	Heat shock protein HSP 90-beta	82.0-84.0 kDa
84.79	Hsp90aa1	P07901	HS90A_MOUSE	Heat shock protein HSP 90-alpha	84.0-86.0 kDa
85.73	Tfrc	Q62351	TFR1_MOUSE	Transferrin receptor protein 1	84.0-86.0 kDa
95.31	Eef2	P58252	EF2_MOUSE	Elongation factor 2	94.0-96.0 kDa
112.98	Atp1a1	Q8VDN2	AT1A1_MOUSE	Sodium/potassium-transporting ATPase subunit alpha-1	112.0-114.0 kDa
116.72	Vcl	Q64727	VINC_MOUSE	Vinculin	116.0-118.0 kDa

Complete molecular-weight bin summary, low to high

All molecular-weight bins are shown. Long Families and Examples / audit families cells wrap and continue across pages.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
0.0-2.0 kDa	61	0	0		routine	. Ighd1-1, . Ighd1-3, . . . , Ighd6-4, , Iglj3p, . Iglj2, . Iglj3, Iglk5
2.0-4.0 kDa	79	0	0		routine	Traj9, Traj25, Traj24, Traj31, Traj41, Trgj2, Traj34, Traj33, Traj4, Traj26, Traj39, Traj6, Traj47, Traj7, Traj42, Traj38, Traj35
4.0-6.0 kDa	41	0	0	Mitochondrial markers	routine	Os14, Gm56820, Carlr, Mtrn1217, , 1700047F07Rik, , Gm58497, Gm49027, Tmsb10, Gm63526, Mrln, Gm14190, . , Tmsb15b2, Tmsb15b1, Tunar
6.0-8.0 kDa	138	0	0	Actin family; Mitochondrial markers	elevated	Gm64043, Mt1, Krtap19-9b, Tmmn5 [Mitochondrial markers], Pln, Smn30, Mt2, Misfa [Mitochondrial markers], Tmmn7 [Mitochondrial markers], Cops9, Krtap20-1, Mt4, Defb46, Rpl39l, Smn11l, Cox14 [Mitochondrial markers], Atp5mk [Mitochondrial markers], Gm13398
8.0-10.0 kDa	238	0	1	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Mitochondrial markers; p53 pathway	elevated	Gm6993, Krtap6-1, Cox8c [Mitochondrial markers], . Clc, Glf2h5, Defb50, Fam236d, Tmem167a, Hsbp1l1 [Heat shock proteins], Hcst, Spag11a, Mzt1, Ndula1, Gm40193, Romo1, Rpl38, Atp5me [Mitochondrial markers]
10.0-12.0 kDa	467	0	1	Actin family; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers	elevated	Dbi, A030005L19Rik, Lce3a, Acdb7, Scgb1b19, Iapp, Gm49835, Lce3f, Mllt11, Lce3d, Gm94, Gm10267, Coxfa4f2 [Mitochondrial markers], S100a6, Scgb1b12, Lce3e, Defa31, Scgb1b3
12.0-14.0 kDa	745	0	5	Actin family; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers; Tubulin family; p53 pathway	high	 , Pex39, Slurp1, Emc6, Dnal4, , Lym7f, Trpc5os, . , Tmem265
14.0-16.0 kDa	436	3	5	Actin family; Apoptosis caspase/BCL; Histone family; Lamin family; Mitochondrial markers; p53 pathway	high	Crebl2, Trbv12-2, H2ac21 [Histone family], Txndc17, Lce1c, Gpha2, Lce1f, Lce1a2, Gabarapl1, H2aj [Histone family], Sncb, H2ac1 [Histone family], Batf, Anp32c, Ly6g6c, , Trav15d-1-dv6d-1, Ly6c2
16.0-18.0 kDa	461	1	4	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; p53 pathway	high	MNcb-2990, Rpl32-ps, Bsp1l, Trappc2l, Sval2 [Actin family], Pthrd1, Vpreb1b, Carhsp1, Ascl4, Slc31a2, 1700029F12Rik, Il9, Jpt1, Rps19, Cnih4, Cstf2, Cst9, Cox5a [Mitochondrial markers]
18.0-20.0 kDa	448	1	8	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Mitochondrial markers; Tubulin family	high	Spaca5, Ube2t, Jchain, I36a, Ncbp2, Czib, Sxl1, Pttg1ip2, Pmp22, Wldc2, Gm16405, Cyrr1, 6030498E09Rik, Spata19 [Mitochondrial markers], Eola1, 2410137M14Rik, Ucn3, Ten1
20.0-22.0 kDa	585	0	19	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	Tmem240, Obp2b, Sat1, Urad, Ssr2, Jpt2, Park7, Itgb3bp, Csn3, Igl2, Faiml, Ssxb13, Reg3d, Faap20, Cenpm, Esm1, Smap, Bcl2a1d [Apoptosis caspase/BCL]
22.0-24.0 kDa	556	0	28	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family	high	Ctf2, Ankub1, Rhoc, Lrrc51, Ctag2l2, Ap3s2, Gsc2, Fldc2, Rcan2, Cbfb, Btf3, Ccdc28b, Commd3, Tceal5, Tmem125, Krtap10-10, Angptl8,
24.0-26.0 kDa	666	0	13	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Lamin family; Mitochondrial markers; p53 pathway	high	Zfand6, Nme5, Rogn1, Yae1d1, Aicda, Ndufs8 [Mitochondrial markers], Coq7 [Mitochondrial markers], Tpd5l2, Cbln2, Tex35, Csprs, Orm3, Mospd1, Muc3a, Evi2a, Cabccoc1, Cldn24, Ms444b
26.0-28.0 kDa	559	0	12	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	Mpig6b, Mdfi, Lat, Malsu1 [Mitochondrial markers], Clns1a, Tmed4, Ai597479, Them4, Mob4, Rnd1, Tirap, Ceacam9, Gm20812, Prl2c1 [Actin family], Zlyve21, Gm20834, Tspan4a, Mzp3
28.0-30.0 kDa	506	1	7	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel	high	Krtap4-27, Crybb1, Hoxb9, Pdc, Rpl8, Hdhd3, Tmt1a3, Tmem33, Wbp2, Amy2a1, Ccdc59, Pip4p2, N4bp2l1, Neur13, Kll15, Tmt1a2, Pcgf3, Tspan3
30.0-32.0 kDa	490	1	10	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Lamin family; Mitochondrial markers	high	Svs3b, Sna12, Gm9758, Rchy1, Fltm2, Mapre1, 2610028H24Rik, Alyreffm10, Fcgr3, Nsa2, Speer4e2, , Pip4p1, Plpbp, , Dhrs2, Dnqj9, Dcun162
32.0-34.0 kDa	545	1	9	AKT pathway; Actin family; Cell-cycle CDK/cyclin; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	Zmat3, Gjl1, C1qtnf1, Oser1, Nkx1-2, Ttpa, D17h653e, Sh2d6, Btbtd19, Dnase1, Mrgprg, Isoc1, Bcdin3d, Plbdl1, Elovf3, Hey1, Ssr1, Eglf8
34.0-36.0 kDa	1722	3	14	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	Or4b1, Cd276, Slc39a3, Spry1, Tmmn40l [Mitochondrial markers], Dhrs1, Or4c127, Ogn, Pex26, Or1j10, Ceacam12, Syp, , Or8d1b, Cdk5r1 [Cell-cycle CDK/cyclin], Or3a4, Katnbl1, Vmn1r63
36.0-38.0 kDa	734	0	12	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	Vmn1r193, V1ra11, Ankrd1, Or2a6f, Or5a15, Frzb, Or9k7, Prss32, Vmn1r84, Tasr122, Or4a27, Vmn1r227, Or14j6, Alad, Gm5655, Srx20, Ammrecr1, Wdr5b
38.0-40.0 kDa	553	0	20	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	Rcn3, Taar8b, Pax4, Borsc6, Cts6, Opcm1, Rnf167, Pltgr2, Rnf39, Gmpr2, Wdr45b [Actin family], Acmsd, Scamp1, Tas2r123, Elmod1, Zfp696, Gimap4, Sl3ga4
40.0-42.0 kDa	551	1	20	Actin family; Cell-cycle CDK/cyclin; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	Ptgdg, Sstr5, Sp6, Plek2, Nans, Cxcr1, Lgals9, Gpr27, Rpl1, Septin3, Spn, Gjd4, Lhx6, Ccr8, Rbm4, Tob1, Pim2, Lrrc69
42.0-44.0 kDa	503	0	24	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	Actb2 [Actin family], Acta2 [Actin family], Foxo3, Arf3, Opn5, Actc1 [Actin family], Septin1, Syne4, Gna11, Rnf34, Yy2, Hsd3b3, Fut9, Mapk12 [MAPK cascade], Sstr4, Ldhalfb, Acta1 [Actin family], Tfrf2
44.0-46.0 kDa	451	0	9	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers	high	Foxh1, Mtg2 [Mitochondrial markers], Amt [Mitochondrial markers], Lhx3, Fnta, Npy1r, Vsig1, Sirpb1a, Lhx9, Mapkapk2 [MAPK cascade], Gna13, Gpr137b, Lef1, Nipa4, Tex9, Mvd, Zfp707, B4gal3
46.0-48.0 kDa	510	1	22	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	Sdc3, Serpina1a, Ccdc71, Aadac12fm2, Mppe1, Cers4, Abrandax1, Pdk2 [Mitochondrial markers], Sec14a, Igsf11, Acnat1, Rtn4r2, Cers3, Scl14c3, Nr2f1, Aziz, Rpl3, A24ad4fm3
48.0-50.0 kDa	430	3	10	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Histone family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family	high	Dph1, Chst9, Rnf150, Kcnj16, Entr1, Krt23, Dok3, Tead4, B02001113Rik, Tbccl [Tubulin family], Tbx19, Bzw1, Foxd4, Ccnb1 [Cell-cycle CDK/cyclin], Fam117a, Cmas, Pnma8a, Bzw2
50.0-52.0 kDa	408	3	8	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	Serpina3f, Ccdc91, Runcd3a, Adss2, Zfp113, LOC102638047, Cmpk2 [Mitochondrial markers], Cdc25c, Tuba8 [Tubulin family], Eef1g, Zfp362, Errf1, Hlx, Wp1f [Actin family], Slc39a8, Slc46a1, Tubb6 [Tubulin family], Tgfb11
52.0-54.0 kDa	469	0	13	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	Chia, Serpinc1, Fezf1, Cpsf7, Zfp874a, Gsdma3, Krtap16-1, Rmrdn3, Tekt4, Wrap73, Altg4c, Srx8, Mettl17 [Mitochondrial markers], Gira2, Slc36a2, Nr5a1, Fam124b, Cables2
54.0-56.0 kDa	473	0	10	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	Zfp810, Tmem200a, Etv4, Nab1, Tmem161a, Pnlppr2, F10, Ltv1, Polr1e, Pnma3, Krt24, Scarb2, Fam114a2, Insm1, Pkmyt1, Klf11, Zfp872, Abtb1
56.0-58.0 kDa	475	1	9	AKT pathway; Actin family; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Mitochondrial markers; Nuclear receptor PPAR	high	Usp14, Irf5, Klc3, Cyp2c70, Cyp2c66, LOC620551, Traf2, Slc18a1, Slc43a3, Rnf217, Cyp2t4, Gom2, Nagpa, Zfp1p3, Matk, Cyp2c68, Slc1a4, Pramel45
58.0-60.0 kDa	378	1	10	Actin family; Cell-cycle CDK/cyclin; ER markers; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	Cct6a, Ccnl2 [Cell-cycle CDK/cyclin], Fto, Cyp19a1, Cyp3a59, Crnn, Agtgf1, Gm10778, Lao1, Kllh33, Cct4, Znf513, Hinfp [Histone family], Sntb1, Zfp273, Slc29a4, Rorc, Vrk2
60.0-62.0 kDa	358	0	7	Actin family; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	Fmo5, Ugt1a9, Slc22a6, Mon1b, Slc16a2, Nxe5, N4bp3, Tmem184c, Ugt1a1, Flvcr2 [Lamin family], Bmpr1a, Vangl1, Hpse, Traf6, Adgrg3, Ugt1a5, Rbm46, Pank1
62.0-64.0 kDa	318	0	4	Actin family; Apoptosis caspase/BCL; Mitochondrial markers; Tubulin family; p53 pathway	elevated	C9, Mgat3, Lmod2, Prkaa2, Coasy, Ntn3, Slc1a2, Pigz, Syt14, Slc22a27, Gpr107, Fbxw11, Dll3, Prf1, Ces1h, Slc22a28, Tll2 [Tubulin family], Ccdc183
64.0-66.0 kDa	274	0	7	Actin family; Cell-cycle CDK/cyclin; Histone family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	elevated	Arlh1, Nr5a2, Zfp870, Rufy4, A330040F15Rik, Slc15a3, Cdkl2 [Cell-cycle CDK/cyclin], Zfp960, Fzd2, Rbm47, Znf354c, Znf879, , Gm6710, Zfp955a, Parbp [Actin family], Slc26a11, Slc13a2
66.0-68.0 kDa	259	2	4	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Lamin family; Mitochondrial markers	high	Gm8113, Zfp345, Zbtb22, Dcaf8, Frsr1, Gga2, Plk5, Lfrn3, Hrob, Rap1gds1, C8a, Slc13a1, Lrrtm3, Sf7, Lsr, Cdk5rap1 [Cell-cycle CDK/cyclin; Mitochondrial markers], Gm35315, Enc1
68.0-70.0 kDa	238	0	7	Actin family; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers	elevated	Cry1, Lrriq4, Nploc4, Cbfa2t3, Ptpn11, Prodh [Mitochondrial markers], Tesk1, Lingo2 [Actin family], Zfp819, Lrrc40, Ankrd13d, Slu7, Etfdh [Mitochondrial markers], Abim2 [Actin family], Dxb1, Hacd2, Sox13, Ache
70.0-72.0 kDa	253	2	2	Actin family; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers	high	Chml, Plk3, Eng, Mbd1, Zfp997, Ncbp3, Nop9, Slc6a4, Slc23a2, Zmym5, Hspa1a [Heat shock proteins], Mak, Gfce, Rufy2, Il18rap, Epm2aip1 [Actin family], Zfp949, Zap70
72.0-74.0 kDa	223	1	4	Actin family; Heat shock proteins; JAK-STAT; Lamin family; Mitochondrial markers	elevated	Mpp4, Entrep3, Arhgap33os, Raver2, Clap100, Gbp4, Gm12258, Mepce, Trim32, Anks3, Rad21, Fem1aa, Abcg4, Nexn, Hdac10, Slfn3, Eif2, Slc9a9
74.0-76.0 kDa	202	1	2	AKT pathway; Actin family; Heat shock proteins; Lamin family; Mitochondrial markers; Tubulin family	elevated	Ckap2, Abcg1, Duxd3, Ptprr, AU018091, Tllf12 [Tubulin family], Hdc, Zbtb33, Sass6, Dytn, Ints9, Extl1, Rnf6, Rpap3, Syn1, Dars2 [Mitochondrial markers], Mmp2, Gtpbp4
76.0-78.0 kDa	224	0	6	AKT pathway; Actin family; Heat shock proteins; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	elevated	Ctbp [Mitochondrial markers], Foxred2, Wdr48, Atf6b, Gramd1c, Mcmcd2, Sytl4, Zfp976, Zfp53 [p53 pathway], Ccdc30, Capn13, Pwpp3a, 4933409G03Rik, Macc1, Pde4c, Ilf70b, Shkbp1, Os9
78.0-80.0 kDa	199	0	6	Actin family; Apoptosis caspase/BCL; Mitochondrial markers; NF-kB/Rel	elevated	Adarb1, Acsl6, Zcchc8, Thop1, Plpl, Fbln1, Tnfaip2, Med25, Slc3a1, Dvl3, Rnf19b, Fndc7, Rasgrp3, Caprin1, Sh3kbp1, Zfp948, Cwh43 [Actin family], Cdh13
80.0-82.0 kDa	200	0	5	AKT pathway; Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel	elevated	Wdr49, Kllh4, Eps8l1, Yme1l1, Zfp111, Gp1ba, Sh3f2, C1ra, Nol10, Eif2b5, Fancb, Tpmn, Slc44a2, Ski, Cpeb4, Lpo, Rsp44a, P3h2
82.0-84.0 kDa	203	1	4	AKT pathway; Actin family; Heat shock proteins; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	Ints10, Map7, Zdhhc8, Cog2, Vps52, Kri1, Ext2, Fign, Capn1, Ddx50, Adarb2, Ngef, Mapk6 [MAPK cascade], Nrg2, Trpv5, Eps8l2, Adgrf4, Tdrd3

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
84.0-86.0 kDa	158	2	3	AKT pathway; Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; Mitochondrial markers; NF-kB/Rel	elevated	Zfp184, Limal1 [Actin family], Dtna, Fam83c, Sytl5, Sox5, Arhgap24, Cdh24, Phtf2, Prss12, Ppp1r3f, Adam11, Dclk1, Glis1, Pcsk1, Zmat1, Haspin, Kank3
86.0-88.0 kDa	189	0	4	Actin family; Cell-cycle CDK/cyclin; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel	elevated	Hgs, Fhip2a [Actin family], Itgb6, Map7d2, Sgip1 [Actin family], Cog7, Mfsd6, Sh3d19, Senp5, Ampd1, Gfm2 [Mitochondrial markers], Ccpg1, Adam25, Adam20, Sec23a, Smurf2, Vps51, Mfn2
88.0-90.0 kDa	128	0	4	Actin family; JAK-STAT; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; p53 pathway	elevated	Vac14, Stat3 [JAK-STAT], Unk, Zw10, Brd2, Pkp2, Cdh11, Zfy2, Prdm4, Lrrc41, Wdr24, L1td1, Mtmr10, Cdhf5, Cdh8, Kat14, Cpt1b, Dclk3
90.0-92.0 kDa	130	0	10	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; JAK-STAT; MAPK cascade; Nuclear receptor PPAR	elevated	Stat5b [JAK-STAT], Zfx, Elnf2, Cpt1c, Adam8, Fyb1, Rasa4, Adams4, Rint1 [Actin family], Uhrf2, Bclaf3 [Apoptosis caspase/BCL], Dthd1, Dna14, Dlgap5, Rrm1, Kank2, Dgkb, Rasgrp1
92.0-94.0 kDa	148	0	5	Actin family; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	Exo1, Taf1c, Avil, Cdc5lrb6, Fam171b, Trim71, Arhgap26, Cdc5lrb7, Adam9, Cdhf3, Col9a1, Txndc16, Tsr1, Pnair, Iqub, Ntkr2, Cdc5lrb, Cdc5lrb1
94.0-96.0 kDa	150	1	2	Actin family; Apoptosis caspase/BCL; Heat shock proteins; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	elevated	Cdhf1, Wdr75, Zbtb10, Vmn2r94, Mical1, Lrrc8a, Exoc6b, Prickle1, Hspa4 [Heat shock proteins], Esys2, Vmn2r13, Nlgn1, Bend3, Tmtc2, Arel1, Ralgsd, Tic7b, Prrm9
96.0-98.0 kDa	191	0	10	Actin family; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel	elevated	Hnmpu1, Polk, Gm61617, Ifl3, Pdcddip [Actin family], Znf629, Sycp21, Map3k12 [MAPK cascade], Kans3, Vmn2r22, Map10, Ttc7a, Vmn2r14, Vmn2r26, Tyro3, Ttk, Axin1, Strip2 [Actin family]
98.0-100.0 kDa	146	0	2	Actin family; Histone family; JAK-STAT; Mitochondrial markers	elevated	Dpp9, Whrn, Trpv4, Esf1, Vmn2r117, Vmn2r109, Lrrc8b, Aco1, Vav1, Atp6v0a2, Dnm2, Vmn2r74, Vmn2r66, Axl, Ar, Il12rb2, Sf3b2, Ccser1
100.0-102.0 kDa	137	0	3	Actin family; Histone family; MAPK cascade; Mitochondrial markers; p53 pathway	elevated	Nfatc2, Cdh4, Esys3, Clca1, Prkd3, Chsy3, Spata31e5, Hk3, Cttna1, Clca3a1, Dlg1, Ermp1, Pla2g4e, Pcdhga3, Mctp2, Psmid2, Dsc3, Ifi206
102.0-104.0 kDa	134	0	6	Actin family; Apoptosis caspase/BCL; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	Lpin1, Ganc, Prkd1, Cip2a, Slc14a2, Pcdhga6, Pcdha10, Mcm2, Snd1, Elf4g2, Hace1, Adgre1, Usp20, Cenpc, Clca4b, Dp8, Grm7, Tcaaf3
104.0-106.0 kDa	116	0	5	Actin family; Cell-cycle CDK/cyclin; JAK-STAT; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family	elevated	Ap2a2, Dgkz, Grk3, Ror1, Thbs3, Lig4, Lgr4, Msh2, Ripor3, Tnp3, Sfswap, Tmtc3, Sik2, Edem3, Gprc6a, , Tbc1d2, Slnf8
106.0-108.0 kDa	93	0	3	Actin family; Apoptosis caspase/BCL; MAPK cascade; Mitochondrial markers	elevated	Bclaf1 [Apoptosis caspase/BCL], Gm61616, Mdga1, Pcsk6, Phc1, Kdm7a, Reck, Rbm19, Ptprr, Plid2, Cherp, Ankle2, Ano5, Rgs3, Clap69, Ankrd24, Gaa, Scart1
108.0-110.0 kDa	95	0	0	Cell-cycle CDK/cyclin	routine	Iqsec1, Slc8a1, Dlgap4, Dp2, Myo19, Chtf18, Als2cl, Thrapp3, Ints4, Gabbr1, , Ipo13, Recqf5, Mical2l, Hk1, Morc1, Usp4, Ints5
110.0-112.0 kDa	83	0	1	Actin family; Histone family; MAPK cascade; Mitochondrial markers	elevated	Emk4, Usp37, Ccdc39, Mib1, Ephb6, Map3k21 [MAPK cascade], Merk, Ern1, Rab3gap1, Ccdc33, Adgrg2, Vps18, Osmr, Tmem132a, Nlrp4f, Srdb1, Zc3h7b, Col6a2
112.0-114.0 kDa	88	1	2	Actin family; Apoptosis caspase/BCL; Histone family; Mitochondrial markers; Nuclear receptor PPAR; p53 pathway	elevated	Syne3, Sipal1, Ppargc1b [Nuclear receptor PPAR], Rad51ap2, Grid1, Atp1a2, Krt78, Grip2 [Actin family], Gtf2, Rbm44, Usp15, Top3a, Ino80d, Pcdh1, Ipo11, Usp35, Uik1, Nlrp4e
114.0-116.0 kDa	80	0	3	Actin family; Apoptosis caspase/BCL	elevated	Atp4a, Crim1, Pdznr4, Smg5, Ano2, Rbm26, Col19a1, Cd101, Tshz2, Crybg3, Pum2, Ugl2, Bicalr [Actin family], Spata31e4, Spata31e3, Card10 [Apoptosis caspase/BCL], Itga9, Sema6a
116.0-118.0 kDa	76	1	0	Actin family; Mitochondrial markers; Tubulin family	elevated	Zfp616, Tdrd5, Taok1, Ipo9, Megf11, Myo1d, Usp38, Epha6, Ccdc57, Clec16a, Tmem132e, Gigyf1 [Actin family], Senp7, Kif17, Agb3, Syde2, Ki, Mrip [Actin family]
118.0-120.0 kDa	90	0	4	AKT pathway; Actin family; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; p53 pathway	elevated	Sk11ip [Actin family], Frmd4b, Kif11, Ubaly1, Pelp1, Specc1, Supg2, Limch1, Ube4a, Kcnh3, U2surp, Nlrp3, Fbh1, Rimb2, Usp11, Bub1b [AKT pathway], Herc4, Vars2 [Mitochondrial markers]
120.0-122.0 kDa	68	0	1	Actin family; Histone family	routine	Znf451, Arhgap30, Zfp729b, Gemin4, Klb, Fhip1a [Actin family], Cebpz, Disl3, Cep131, Sema5b, Kdm4a, Radil, Slc12a1, Adcy4, Cdc4r3, Sretf1, Slc12a4, Usp48
122.0-124.0 kDa	66	0	1	Actin family; JAK-STAT	routine	Grip1 [Actin family], Fmn1l, Trhde, Pde3b, Itga6, Tdrd7, Ubrn1, Dsg2, Slc4a8, Kif1c, Zswim4, Zeb1, Fnip2 [Actin family], Gucy2d, Gucy2g, Lifr, Iqch, Usp8
124.0-126.0 kDa	51	0	2	Actin family; p53 pathway	elevated	Ranbp17, Wwc1, Tacc2, Spata31e1, Amer1, Slc4a5, Sh3pxd2a, Wdhdl1, Adnp, R3hdm1, Wdr72, Pik3ca, Srgap3, Gucy2, Specc1l, Agap2, Pde3a, Zcchc2
126.0-128.0 kDa	58	0	2	Actin family	elevated	Pcdh19, Nek10, Catsperb, Pcdh17, Agbl1, Bmp2k, Btdb7, Lats1, Slc12a5, Oas3, Hps5, Nrip1 [Actin family], Pik3cg, Sin3b, Otoa, Atp13a2, Znf654, Rtn4
128.0-130.0 kDa	64	0	1	Actin family; Lamin family; Mitochondrial markers	elevated	Mybtp1, Abl2, Polr1b, Trappc9, Cfpap7, Itgal, Dhx37, Erc1 [Actin family], Aebp1, Daam2, Rbm6, Trappc11, Ppip5k2, Trpa1, Usp7, Ccdc150, Cep128, Dhx34
130.0-132.0 kDa	62	0	2	Actin family; Apoptosis caspase/BCL; JAK-STAT; Lamin family	elevated	Tbcd8d, Hpk3 [Actin family], Vwa3a, Rbm20, Fnip1 [Actin family], Fbf1, Irs4, Lamc2 [Lamin family], Ppp2r3a, Tecpr1, Sgo2a, BC034090, Adcy6, Kcnt2, Znf687, Cacna2d2, Spata31g1, Zbed4
132.0-134.0 kDa	62	0	1	Actin family; Apoptosis caspase/BCL; JAK-STAT; Tubulin family	elevated	Camta2, Grid2ip, Ccar1 [Apoptosis caspase/BCL], Dnaaf9, Clip, Sorbs2, Catsperg1, Ripor1 [Actin family], Nhsl2, Atp13a1, Tref1, Wnk4, Tul4 [Tubulin family], Atp2b2, Smarcc2, Wwc2, Kdm2a, Fndc3b
134.0-136.0 kDa	68	0	4	Actin family; Apoptosis caspase/BCL; NF-kB/Rel	elevated	Ctcl, Card11 [Apoptosis caspase/BCL], Wapl, Mn1, Washc5, Dgkh, Zbtb38, Nlrp1b, Jag1, Lars1, Smc2, Nlrp1a, Atp2b3, Ltbp3, Mtu51, Kcnma1, Gpr158, Rbp3
136.0-138.0 kDa	58	0	1	Actin family; Histone family; Lamin family; Mitochondrial markers; Tubulin family	elevated	Mmn1, Af3, Alpk1, Pcdh9, Gse1, Adams16, Cand1, Nphs1, Spem3, Washc4, Per1, Cgn, Rapgef1, Atxn2, Nid1, Zeb2, Dhx30, Nek1
138.0-140.0 kDa	49	0	0	Actin family; Histone family; NF-kB/Rel	routine	Dzfp3, Col1a1, Ehm2 [Histone family], Mecom [Histone family], Impg2, Kcnt1, Zbtb40, Nckap5l, Csppl1, Plcb1, Copa, Rti9, Phka2, Nrcam, Inf2, Erbb2, Ptprr, Atf7ip [Actin family]
140.0-142.0 kDa	49	0	0	Actin family; Histone family; MAPK cascade; Tubulin family	routine	Adcy8, Afif2, Vars1, Sall1, Abcb4, Col15a1, Mms22l, Map4k4 [MAPK cascade], Dhx38, Mybpc3, Abcb1a, Hmgxb3, Ithf6, Clasp2, Vwa3b, Ube2o, Cntnap3 [Actin family], Prdm16 [Histone family]
142.0-144.0 kDa	28	0	1	Actin family; Cell-cycle CDK/cyclin; MAPK cascade	routine	Tbcd1d, Arid4a, Gm382, Arhgap29, Dhx8, Sympk, Cdhf2, Ppfia1, Ambra1, Npc1, Cracd [Actin family], Grin2d, Tbcd1d9, Map3k6 [MAPK cascade], Sorbs1, Bcr, Rrp12, Adgra2
144.0-146.0 kDa	48	0	2	Actin family; Histone family	elevated	Sf1, Rasgrf1, Myrn, Mast3, Sik36, Cntnap5b [Actin family], Iars1, Plekha5, Ppp19a, Stag1, Smc1b, Setdb1 [Histone family], Rasa2, Aak, Dennd5b, Plas, Ssc5d, Adgra3
146.0-148.0 kDa	43	0	2	Actin family; MAPK cascade; Tubulin family	elevated	Spata31f1b, Spata31f1c, Spata31f1d, Spag9 [Actin family], Usp42, Cntnap5c [Actin family], Usp31, Map3k19 [MAPK cascade], Xdh, Spata31f1a, Simc1 [Actin family], Dennd5a, Aox1, Clsnp, Abcb11, Erbb4, Smc4, Mroh5
148.0-150.0 kDa	44	0	2	Actin family; Apoptosis caspase/BCL; MAPK cascade	elevated	Trim66, Ptpnm2, Tbcd132, Cluh, Myo3b, Cgnl1, Cntnap2 [Actin family], Syngap1, Kank1, Aox4, Znf541, Zfp644, Disp3, Plekhg3, Clap251, Ttc21a, Rnf123, Cngb1
150.0-152.0 kDa	36	0	0	Actin family	routine	Kif16b, Phldb1, Kif24, Nphp3, Pds5a, Trnk [Actin family], Carmil3, Kdr, Pnpla7, Caskin1, Mst1r, Usp19, Sik3, Acin1, Ttc21b, Ttc41, Sos1, Lmk3
152.0-154.0 kDa	30	0	1	Actin family; Histone family	routine	Nup153, Tns2, Mybbp1a, Palld, Npat, Nsd2 [Histone family], Cdc171, Cpd, Zfyve9, Atrn1 [Actin family], Edc4, Rab3gap2, Ktn1, Pik3r4, Frmpd2, Zmym3, Flt4, Cemip
154.0-156.0 kDa	30	0	1	Actin family; MAPK cascade	routine	Spata31e2, Kdm6a, CEFIP [Actin family], Map3k5 [MAPK cascade], Zmyrn2, Cux2, Kiaa0232, Pogz, Iqcn, Tecpr2, Nup155, Ylpm1, Fgd6, Gm60204, Rpap1, Arhgap31, Adams13, Fam135b
156.0-158.0 kDa	19	0	1	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers	routine	Med23, Cntnap1 [Actin family], Mia2, Rprd2, Lrprr [Mitochondrial markers], Bcl9 [Apoptosis caspase/BCL], Nacad, Btp3a, Ncoa1, Gm1043, Atp7b, Wn, Erbin, Nph4, Setd5 [Histone family], Usp47, Rfx7, Scaper [Cell-cycle CDK/cyclin]
158.0-160.0 kDa	23	0	2	Actin family; Cell-cycle CDK/cyclin	routine	Atrn [Actin family], Rock1, Ssh2, Nup160, Atp10d, Dcc, Blm, Ncoa2, Synj2, Naip1, Drosha, Ric1, Kif27, Shank2, Ccnb3 [Cell-cycle CDK/cyclin], Ankrd48, Ptch1, Naip7
160.0-162.0 kDa	41	0	2	Actin family; Histone family; MAPK cascade	elevated	Kif15, Mbd5, Nos1, Smg6, Lmk2, Rock2, Col4a1, Cntln, Uaca, Cpsf1, Cep162, Trappc8, Prrc2b, Eea1, Med14, Nsd3 [Histone family], Ythdc2, Abcc5
162.0-164.0 kDa	27	0	2	Cell-cycle CDK/cyclin; MAPK cascade	routine	Scaf11, Arap1, Fyco1, A2m1l, Znf608, Mug2, Gapvd1, Ptprr, Mlh3, Tex14, Cep164, Rad54f2, Asx1, Mapkbp1 [MAPK cascade], Neo1, Ptprr, Rims1, Shroom4
164.0-166.0 kDa	38	0	1	Actin family; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers	routine	Col4a6, Naip2, Col4a4, Plekhg1, Ptprr, Plch2, Camspap2, A2m, Pds5b, Plb1, Cdk13 [Cell-cycle CDK/cyclin], Cps1 [Mitochondrial markers], Shroom2, Megf6, Alad2b, Myom2, Dennd4b, Abcc6
166.0-168.0 kDa	26	0	0		routine	Nrxn1, Rapgef2, Atp10b, Adgrl2, Gemin5, Ccdc88b, Kcp, Zfyve16, Talpid3, Fam135a, Lrrc7, Fam193a, Mrc2, Nexmif, Med1, Golga3, Wdr62, Lrrc9
168.0-170.0 kDa	18	0	0		routine	Shoc1, Ahdc1, 493040710Rik, Sli2, Atp10a, Topbp1, Dcaf1, Resf1, Tut7, Frmpd1, Abcc3, Clasp1, Nhsl1, Ncapd3, Tulp4, Ccdc18, Arap3, Adgrb2
170.0-172.0 kDa	20	0	1	AKT pathway; Actin family	routine	Eprs1, Disp1, Agr, Arhgap35, Pla2r1, Baz1b [AKT pathway], Cep170b, Mast1, Dip2c [Actin family], Adgrl3, Dip2b [Actin family], Abcc1, Adgrb3, Col11a2, Pik3c2g, Urb2, Gli3, Rere
172.0-174.0 kDa	24	0	0	Lamin family	routine	Sycp2, Cdc42bpg, Trpm2, Lamc3 [Lamin family], Arhgef12, Slx4, Zmyrn4, Arhgap5, Synj1, Pcf11, Ugg12, Top2a, Rims2, Rrbp1, Setpb1, Tnn, Synn, Nwd1
174.0-176.0 kDa	19	0	0	Nuclear receptor PPAR	routine	Zfp831, Abcc9, Agl, Duox2, Elf4g3, Alk, Nisch, Cep170, Pprc1 [Nuclear receptor PPAR], Madd, Kdm5c, Kdm5b, Fhod3, Tarbp1, Col24a1, Elcabc, Tmem131l, Trnc6c
176.0-178.0 kDa	17	0	0	Lamin family	routine	Elf4g1, Hecw2, Kdm6b, Robo1, Ugg1t, Plxnc1, Ino80, Usp54, Arhgef5, Kdm5d, Duox1, Gm32742, Dnmbp, Rimb3, Lamc1 [Lamin family], Tiam1, Adams12
178.0-180.0 kDa	15	0	1	Actin family; MAPK cascade	routine	Abcc8, Baz1a, Lthp4, Cttnbp2 [Actin family], Lrp5, Cfpap74, Peg3, Nhs [Actin family], Rapgef6, Thsd7b, Hecw1, Abi3bp, Katnip [Actin family], Scrib, Map3k4 [MAPK cascade]
180.0-182.0 kDa	21	0	0	Actin family	routine	Erich3, Alpk3, Samd9l, Smarca2 [Actin family], Lrp6, Depdc5, Mroh2b, Iqgap2, Ankrd26, Akap12, Col11a1, Kiaa1210, Pappa, Tasor, Smarca4 [Actin family], Prex2, Usp32, Mroh1
182.0-184.0 kDa	16	0	1		routine	Col18a1, Adams7, Rb1cc1, Phipp1, Als2, Thoc2l, Thoc2, Siglec1, Abca8b, Abca9, Mtlc2, Dnm1t, Abca6, Thsd7a, Trpm1, Col5a2
184.0-186.0 kDa	22	0	1		routine	Phrf1, Rreb1, Pik3c2b, Abca8a, Wiz, Camta1, Tut4, Mdc1, N4bp2, Myo3a, Ehbpl11, Nrxn2, Prex1, Ccdc7a, Sphkap, Ptpn23, Iqgap3, Shank3
186.0-188.0 kDa	16	0	0	Histone family	routine	Setd1a [Histone family], Kif21b, Col27a1, Adcy10, Kif14, C3, Elf2ak4, Kif21a, Crocc2, Ltbp1, Ptprr, Pbrm1, Wdr97, Prdm2, Cc2d2a, Plch1

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
188.0-190.0 kDa	8	0	0		routine	Adgb, Iqgap1, C5, Itsn2, Mon2, Sipai12, Adamts13,
190.0-192.0 kDa	18	0	1		routine	Mroh2a, Bcor1, Arhgef28, Mast2, Abca15, Pik3c2a, Btbd8, Peak1, Casz1, Kndc1, Kdm3b, Shprh, Cltc, Rictor, Kif1a, Lrriq1, Tnrc6b, Abca3
192.0-194.0 kDa	16	0	0		routine	Adamts11, Marf1, Scn7a, Kdm5a, Abca16, Abca14, Cul7, Tiam2, Bcor, C4a, C4b, Col14a1, Arap2, Cfpap43, Unc13a, Wnk3
194.0-196.0 kDa	19	0	0		routine	Polr1a, Lmo7, Itsn1, Xrn1, Qser1, FAM186A, Gcc2, Tjp1, Cdc42bpb, Tet3, Togaram1, Sipai13, Frmpd3, Cdc42bpa, Nrap, Kif13a, Ltbp2, Ccdc180
196.0-198.0 kDa	22	0	0	Actin family; Lamin family; Tubulin family	routine	Abca17, Klf26a, Strc, Chd1, Cep152, Lamb2 [Lamin family], Xirp1 [Actin family], Nup188, Zfp646, Sipai11, Zswim8, Lamb1 [Lamin family], Tubgcp6 [Tubulin family], Nup98, Ly75, Nwd2, Trpm3, Ift172
198.0-200.0 kDa	14	0	1	Actin family	routine	Patj, Brca1, Pappa2, Ltn1, Focad, Rtf1, Supt6h, Map2, Ubr2, Kidins220 [Actin family], Rbbp6, Rp111, Fndc1, Tspoap1
200.0-202.0 kDa	9	0	0	Lamin family	routine	Ubr1, Nalcn, Tanc1, Tns1, Scn11a, Virma, Rai1, Itgb4, Lama4 [Lamin family]
202.0-204.0 kDa	14	0	0		routine	Tsc2, Arlgef2, Nav1, Myo5c, Zfp236, Brwd3, Wdr90, Nynrin, Tnrc6a, Kif20b, Zc3h13, Parp14, Atad5, Ecpas
204.0-206.0 kDa	9	0	1	Cell-cycle CDK/cyclin; Histone family; p53 pathway	routine	Ppl, Kif1b, Nup210 [p53 pathway], Ank1, Kmt2e [Histone family], Tmem131, Kif13b, Zgrf1, Cdk5rap2 [Cell-cycle CDK/cyclin]
206.0-208.0 kDa	13	0	0	Actin family	routine	Tro, Ankrd31, Znf142, Plxnb2, Afdn, Notch4, Phip [Actin family], Gb1, Btaf1, Nes, Agrn, Pdcd11, Plxna3
208.0-210.0 kDa	18	0	0	Histone family; p53 pathway	routine	Tchh, Ticrr, Plxnb3, Arlgef1, Kat6b [Histone family], Sbf1, Nup210 [p53 pathway], Akap11, Scn4a, Znf106, Sis, Nckap5, Ckap65, Kiaa1549, Pcsk5, Dennd4a, Baz2a, Fer16
210.0-212.0 kDa	21	0	0		routine	Ralgapa2, Cacna1s, Hcfc1, Sbf2, Myo5b, Chd2, Atg2a, Plxna1, Psme4, Myo16, Dennd4c, Ptipr1, Plxna2, Cfpap44, Plknd1, Dock2, Nlr5, Prr12
212.0-214.0 kDa	14	0	1	p53 pathway	routine	Adamts20, Tet2, Trpm7, Plxna4, Tp53bp1 [p53 pathway], Ubr3, Mylk, Nup214 [p53 pathway], Skt, Ttc6, Vwa8, Greb1l, Mia3, Mtdcl
214.0-216.0 kDa	17	0	0	Histone family; p53 pathway	routine	Ank3, Muc12, Dlg5, Ptprd, Taf1, Dock5, Pcdh15, Shroom3, Dock1, Setd1b [Histone family], Myo5a, Tcf20, Adamts9, Prr14l, Arhgap21 [p53 pathway], Ccdc88a, Anapc1
216.0-218.0 kDa	11	0	0		routine	Parp4, Dicer1, Greb1, Polr2a, Thada, Mcm3ap, Rlf, Exph5, Chd4, Lct, Emi6
218.0-220.0 kDa	10	0	1	Actin family; Apoptosis caspase/BCL	routine	Mical2 [Actin family], Znf638, Mpdz, Znf1x1, Brd10, Casp8ap2 [Apoptosis caspase/BCL], Ak9, Ncoa6, Helz, Heatr5a
220.0-222.0 kDa	11	0	0		routine	Emi5, Tanc2, Naca, Scn10a, Scn3a, Myh7b, Pcnx3, Arhgef17, Myh15, Cacna1f, Acan
222.0-224.0 kDa	16	0	0	Actin family	routine	Dscam, Chd5, Myh8, Myh4, Myh7, Tet1, Myh2, Myh1, Triobp [Actin family], Crybg1, Myh13, Myh6, Mical3 [Actin family], Myh3, Fer1l4, Ttc3
224.0-226.0 kDa	14	0	0	Histone family	routine	Trip12, Dscaml1, Heatr5b, Ptipr, Kat6a [Histone family], Zfc3h1, Scn8a, Lrrk1, Kif26b, Phf3, Ckap5, Smchd1, Scn9a, D430041D05Rik
226.0-228.0 kDa	13	0	0		routine	Fancm, Shank1, Myh9, Trip11, Ccdc88c, Dock4, Dmbt1, Crocc, Myh11, Otof, Wnk2, Scn5a, Scn2a
228.0-230.0 kDa	7	0	0		routine	Myh14, Scn1a, Pcm1, Myh10, Prrc2a, Ralgapa1, Arhgap32
230.0-232.0 kDa	4	0	0		routine	Plxnb1, Atg2b, Gm19410, Tnc
232.0-234.0 kDa	11	0	0		routine	Evpl, Chd3, Myo18a, Trpm6, Dock3, Ankrd12, Esp1l, Nup205, Dock6, Alpk2, Myof
234.0-236.0 kDa	9	0	0	Actin family	routine	Baz2b, Pcnx2, Rp1, Frem3, Dock9, Cit [Actin family], Numa1, Loxhd1, Fer1l5
236.0-238.0 kDa	11	0	0		routine	Malr1d, Pkd113, Pikfyve, Abca7, Pl4ka, Myo10, Gtf3c1, Usf3, Dock11, Dysf, Tdrf6
238.0-240.0 kDa	6	0	0		routine	Knl1, Med13, Myo9b, Dock8, Tecta, Sdk2
240.0-242.0 kDa	8	0	0		routine	Arlgef3, Cacna1c, Sdk1, Myo7b, Pkdrej, Med12l, Dock7, Med13l
242.0-244.0 kDa	12	0	0		routine	Heatr1, Arid1a, Ice1, Arid1b, Cacna1i, Armcx4, Apc2, Svll, Cabin1, Cad, Asxl3, Nin
244.0-246.0 kDa	7	0	0		routine	Notch3, Frem1, Snmp200, Med12, Ints1, Sptb, Zfp407
246.0-248.0 kDa	8	0	0		routine	Col6a6, Znf318, Cacna1d, Sor1, Dido1, F5, Hrrn, Ahctf1
248.0-250.0 kDa	5	0	0		routine	Rttm, Gpr179, Gon4l, Dock10, Unc13c
250.0-252.0 kDa	6	0	0		routine	Kntc1, Ascc3, Pde4dip, Col6a4, Wnk1, Flg2
252.0-254.0 kDa	5	0	0		routine	Nav3, Cspg4, Hivp3, Tin2, Abca1
254.0-256.0 kDa	9	0	0	Heat shock proteins	routine	Sec16a, Akap6, Ptipr1, Btp2, Urb1, Myo7a, Plce1, Dnajc13 [Heat shock proteins], C2cd3
256.0-258.0 kDa	6	0	0		routine	Ptipr, Potefam1, Cacna1e, Dop1b, Golga4, Catspert
258.0-260.0 kDa	3	0	0		routine	Cic, Pcnx1, Brwd1
260.0-262.0 kDa	5	0	0		routine	Abca4, Otogl, Cacna1b, Nav2, Ros1
262.0-264.0 kDa	5	0	0	Histone family	routine	Cacna1h, Pole, Cacna1g, Spag17, Ep300 [Histone family]
264.0-266.0 kDa	8	0	0	Histone family	routine	Tasor2, Mgam2-ps, Acaca, Crebbp [Histone family], Notch2, Son, Cspg4b, Nbas
266.0-268.0 kDa	6	0	0		routine	F8, Rlf1, Hivp2, Cnot1, Ttc28, Cacna1a
268.0-270.0 kDa	5	0	0		routine	Cntrf, Dop1a, Ankhd1, Ncor2, Tin1
270.0-272.0 kDa	8	0	0		routine	Map1b, Ptpn13, Abca2, Ncor1, Bdp1, Notch1, Sptbn2, Pkd1l2
272.0-274.0 kDa	9	0	0		routine	Fasn, Fn1, Cep295, Zfhx2, Prpf8, Zdbf2, Igl2r, Spg11, Tpr
274.0-276.0 kDa	4	0	0		routine	Ankrd17, Sptbn1, Fcgbp, Acacb
276.0-278.0 kDa	5	0	0	Lamin family	routine	Stab1, Cep192, Cep250, Stab2, Flnb [Lamin family]
278.0-280.0 kDa	2	0	0		routine	Atrx, Spta1
280.0-282.0 kDa	5	0	0	Lamin family	routine	Fcgbpl1, Polq, Gvin1, Flna [Lamin family], Jmjd1c
282.0-284.0 kDa	4	0	0		routine	Cul9, Bahcc1, Znf462, Zfyve26
284.0-286.0 kDa	6	0	0	Histone family	routine	Mast4, Nsd1 [Histone family], Sptan1, Lrrk2, Igsf10, Setd2 [Histone family]
286.0-288.0 kDa	1	0	0		routine	Cenpe
288.0-290.0 kDa	7	0	1		routine	Hivp1, Col6a3, Mtor, Sptbn4, Myo18b, Cep290, Col6a5
290.0-292.0 kDa	9	0	0	Lamin family	routine	Hectd1, Usp9x, Unc79, Epg5, Pkd11, Chd8, Flnc [Lamin family], Tep1, Trrap
292.0-294.0 kDa	5	0	0		routine	Piezo1, Myo9a, Usp9y, Abca12, Gcn1
294.0-296.0 kDa	5	0	0	Histone family	routine	Usp24, Gm9195, Kmt2b [Histone family], Srm2, Col7a1
296.0-298.0 kDa	4	0	0		routine	Ankrd11, Pdzd2, Megf8, Setx
300.0-302.0 kDa	5	0	0		routine	Map1a, Atr, Muc6, Zfp292, Nbeal2
302.0-304.0 kDa	4	0	0		routine	Clap46, Tenm3, Igfb1, Akap13
304.0-306.0 kDa	4	0	0		routine	Itpr3, Tg, Chd6, Tenm1
306.0-308.0 kDa	4	0	0		routine	Nbeal1, Tenm2, Itpr2, Tnrc18
308.0-310.0 kDa	3	0	0		routine	Ubr5, Tenm4, Vwf
310.0-312.0 kDa	3	0	0		routine	Prrc2c, Apc, Tex15
312.0-314.0 kDa	4	0	0		routine	Fbn1, Itpr1, Otof, Fbn2
314.0-316.0 kDa	1	0	0		routine	Nipbl
316.0-318.0 kDa	3	0	0		routine	Celsr2, Lrba, Utp20
318.0-320.0 kDa	1	0	0		routine	Nf1
322.0-324.0 kDa	1	0	0		routine	Chd9
324.0-326.0 kDa	1	0	0		routine	Piezo2
326.0-328.0 kDa	2	0	0		routine	Nbea, Bod1l
328.0-330.0 kDa	3	0	0		routine	Zzef1, Mga, Pcnt
330.0-332.0 kDa	3	0	0	Histone family	routine	Celsr1, Ash1 [Histone family], Helz2
332.0-334.0 kDa	3	0	0		routine	Dsp, Myo15b, Bptf
334.0-336.0 kDa	2	0	0		routine	Chd7, Adgrg4
336.0-338.0 kDa	4	0	0		routine	Dmx1, Kalm, Ep400, Fryl
338.0-340.0 kDa	4	0	0	Lamin family	routine	Lama1 [Lamin family], Dmx2, Fry, Prune2
340.0-342.0 kDa	2	0	0		routine	Col12a1, Ranbp2
342.0-344.0 kDa	3	0	0	Lamin family	routine	Cenpf, Trank1, Lama2 [Lamin family]

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
344.0-346.0 kDa	1	0	0		routine	Htt
346.0-348.0 kDa	3	0	0		routine	Dchs1, Cep350, Trio
348.0-350.0 kDa	2	0	1		routine	Srcap, Atm
350.0-352.0 kDa	3	0	0		routine	Frem2, Rev3l, Mki67
352.0-354.0 kDa	1	0	0		routine	Cfap54
354.0-356.0 kDa	2	0	0		routine	Wdly4, Speg
356.0-358.0 kDa	1	0	0		routine	Dnah12
358.0-360.0 kDa	3	0	0		routine	Celsr3, Cplane1, Vps13a
360.0-362.0 kDa	2	0	0		routine	Alms1, Cfap47
364.0-366.0 kDa	2	0	0		routine	Aspm, Dchs2
366.0-368.0 kDa	3	0	0	Lamin family	routine	Lama3 [Lamin family], Lrrc37a, Vcan
368.0-370.0 kDa	3	0	0		routine	Muc4, Ahnak2, Cdh23
370.0-372.0 kDa	3	0	0		routine	Golgb1, Unc80, Brca2
372.0-374.0 kDa	1	0	0		routine	Muc5ac
376.0-378.0 kDa	1	0	0		routine	Szt2
386.0-388.0 kDa	3	0	1	NF-kB/Rel	routine	Svep1, Reln [NF-kB/Rel], Csmc1
390.0-392.0 kDa	1	0	0		routine	Gm58671
392.0-394.0 kDa	5	0	0		routine	Zfhx4, Wdly3, Csmc2, Utrn, Spata31h-ps1
394.0-396.0 kDa	1	0	0		routine	Myo15a
398.0-400.0 kDa	3	0	0	Actin family; Heat shock proteins	routine	Hspg2 [Heat shock proteins], Spen [Actin family], Cubn
402.0-404.0 kDa	1	0	0		routine	Zfp469
404.0-406.0 kDa	3	0	0	Lamin family	routine	Lama5 [Lamin family], Lrrc37, Csmc3
406.0-408.0 kDa	1	0	0		routine	Zfhx3
408.0-410.0 kDa	2	0	0		routine	Usp34, Smg1
412.0-414.0 kDa	2	0	0		routine	Mgam, Cmya5
414.0-416.0 kDa	1	0	0		routine	Sptbn5
418.0-420.0 kDa	1	0	0		routine	Bsn
420.0-422.0 kDa	1	0	0		routine	Vps13c
424.0-426.0 kDa	2	0	0		routine	Lyst, Dmd
426.0-428.0 kDa	1	0	0		routine	Ank2
428.0-430.0 kDa	2	0	0	Actin family; Histone family	routine	Xirp2 [Actin family], Kmt2a [Histone family]
434.0-436.0 kDa	1	0	0		routine	Akap9
442.0-444.0 kDa	1	0	0		routine	Fras1
444.0-446.0 kDa	2	0	0		routine	Vps13b, Pkhd1
460.0-462.0 kDa	2	0	0		routine	Dnah7a, Dnah7b
462.0-464.0 kDa	2	0	0		routine	Dnah7c, Muc17
464.0-466.0 kDa	1	0	0		routine	Pkhd11
466.0-468.0 kDa	2	0	0		routine	Pkd1, Dnah3
470.0-472.0 kDa	1	0	1		routine	Prkdc
474.0-476.0 kDa	2	0	0		routine	Tnxb, Dnah6
478.0-480.0 kDa	1	0	0		routine	Muc2
480.0-482.0 kDa	1	0	0		routine	Fat2
482.0-484.0 kDa	1	0	0		routine	Huwe1
484.0-486.0 kDa	1	0	0		routine	Hectd4
486.0-488.0 kDa	2	0	0		routine	Dnah1, Vps13d
488.0-490.0 kDa	1	0	0		routine	Gm57895
490.0-492.0 kDa	1	0	0		routine	Gm57894
492.0-494.0 kDa	1	0	0		routine	Dync2h1
500.0-502.0 kDa	1	0	0		routine	Stard9
502.0-504.0 kDa	1	0	0		routine	Fat3
504.0-506.0 kDa	1	0	0		routine	Lrp1
506.0-508.0 kDa	1	0	0		routine	Fat1
508.0-510.0 kDa	1	0	0		routine	Apob
510.0-512.0 kDa	2	0	0		routine	Dnah2, Dnah17
512.0-514.0 kDa	2	0	0		routine	Dnah9, Lrp1b
514.0-516.0 kDa	1	0	0		routine	Muc5b
516.0-518.0 kDa	1	0	0		routine	Dnah11
518.0-520.0 kDa	1	0	0		routine	Lrp2
520.0-522.0 kDa	2	0	0		routine	Sacs, Mycbp2
522.0-524.0 kDa	1	0	0		routine	Dnah10
526.0-528.0 kDa	2	0	0		routine	Herc2, Dnah5
528.0-530.0 kDa	1	0	0		routine	Dnah14
532.0-534.0 kDa	3	0	0		routine	Dync1h1, Birc6, Herc1
534.0-536.0 kDa	3	0	0		routine	Plec, Gm58669, Sspo
536.0-538.0 kDa	1	0	0		routine	Dnhd1
540.0-542.0 kDa	3	0	0	Histone family	routine	Kmt2c [Histone family], Fat4, Dnah8
546.0-548.0 kDa	1	0	0		routine	Hmnc2
550.0-552.0 kDa	3	0	0		routine	Pclo, Ryr3, Spata31h1
554.0-556.0 kDa	1	0	0		routine	Bltp1
564.0-566.0 kDa	2	0	0		routine	Ryr2, Ryr1
568.0-570.0 kDa	3	0	0		routine	Gm57885, Abca13, Ush2A
572.0-574.0 kDa	1	0	0		routine	Ubr4
578.0-580.0 kDa	1	0	0		routine	Zan
580.0-582.0 kDa	1	0	0		routine	Hydin
584.0-586.0 kDa	1	0	0		routine	Rnf213
600.0-602.0 kDa	1	0	0	Histone family	routine	Kmt2d [Histone family]
604.0-606.0 kDa	1	0	0		routine	Ahnak
610.0-612.0 kDa	1	0	0		routine	Hmnc1
628.0-630.0 kDa	1	0	0		routine	Mdn1
686.0-688.0 kDa	1	0	0		routine	Adgrv1
692.0-694.0 kDa	1	0	0		routine	Muc19
700.0-702.0 kDa	1	0	0		routine	Gm58672

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
722.0-724.0 kDa	1	0	0		routine	Eppk1
746.0-748.0 kDa	1	0	0		routine	Ccdc168
782.0-784.0 kDa	1	0	0		routine	Syne2
784.0-786.0 kDa	1	0	0	Actin family	routine	Fsip2 [Actin family]
792.0-794.0 kDa	3	0	0		routine	Gm58665, Gm58676, Gm58666
828.0-830.0 kDa	1	0	0		routine	Neb
830.0-832.0 kDa	1	0	0	Actin family	routine	Macf1 [Actin family]
832.0-834.0 kDa	1	0	0		routine	Gm58662
834.0-836.0 kDa	1	0	0		routine	Dst
870.0-872.0 kDa	1	0	0		routine	Gm61593
908.0-910.0 kDa	1	0	0		routine	Muc16
966.0-968.0 kDa	1	0	0		routine	Obscn
1008.0-1010.0 kDa	1	0	0		routine	Syne1
3906.0-3908.0 kDa	1	0	0		routine	Ttn

Proteins sorted from lowest to highest molecular weight

The PDF shows the first 120 proteins. The complete table is in proteome_wb_integrity_atlas.csv.

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
0.72		P38639	Unknown protein from 2D-PAGE of fibroblasts (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.75	Ighd1-1	A0A0G2JDD8	Immunoglobulin heavy diversity 1-1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.77		P38642	Unknown protein from 2D-PAGE of fibroblasts (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.79	Ighd1-3	A0A1Y7VHD7	Predicted gene, 16968 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.82		P38644	Unknown protein from 2D-PAGE of fibroblasts (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.84		P38641	Unknown protein from 2D-PAGE of fibroblasts (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.89		P99031	Unknown protein from spot 2D-0014LD of 2D-PAGE of liver tissue (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.97	Ighd6-4	A0A0G2JDD5	Immunoglobulin heavy diversity 5-7 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
0.99		P38640	Unknown protein from 2D-PAGE of fibroblasts (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.08		P83853	Vam6-like protein (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.10		P99032	Unknown protein from spot 2D-0014M9 of 2D-PAGE of liver tissue (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.11		P38643	Unknown protein from 2D-PAGE of fibroblasts (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.17	Iglj3p	A0A0G2JFS3	Immunoglobulin lambda joining 3 pseudogene (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.19		P83854	Putative G protein-coupled receptor (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.24	Iglj2	A0A0G2JDZ1	Immunoglobulin lambda joining 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.25		A0ABJ3HP72	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.27	Iglj3	A0A0G2JDG5	Immunoglobulin lambda joining 3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.28	Igkj5	A0A0G2JD11	Immunoglobulin kappa joining 5 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.28	Iglj1	A0A0G2JE61	Immunoglobulin lambda joining 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.31	Igkj2	A0A0G2JDG2	Immunoglobulin kappa joining 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.33	Igkj4	A0A0G2JFT5	Immunoglobulin kappa joining 4 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.34	Igkj1	A0A0G2JEZ6	Immunoglobulin kappa joining 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.38	Igkj3	A0A0G2JGJ9	Immunoglobulin kappa joining 3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.42	Iglj4	A0A0G2JEG4	Immunoglobulin lambda joining 4 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.48		A0ABJ3HKN3	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.64	Trtbj1-2	A0A0A6YWU1	T cell receptor beta joining 1-2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.67	Trtbj1-1	A0A0A6YVY0	T cell receptor beta joining 1-1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.69	Ighj3	A0A0A6YWW9	Immunoglobulin heavy joining 3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.71	Trtbj1-3	A0A0A6YXA0	T cell receptor beta joining 1-3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.71	Trtbj2-3	A0A0B4J1P6	T cell receptor beta joining 2-3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.72	Trtbj1-5	A0A0A6YX16	T cell receptor beta joining 1-5 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.72	Trdj1	A0A0G2JG53	T cell receptor delta joining 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.72		A0ABJ3HKK6	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.72	Ighj2	A0A0A6YWJ8	Immunoglobulin heavy joining 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.73	Trtbj2-2	A0A0A6YX08	T cell receptor beta joining 2-2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.73	Trtbj2-4	A0A0B4J1P9	T cell receptor beta joining 2-4 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.73	Trtbj2-7	A0A0A6YXH6	T cell receptor beta joining 2-7 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.74	Trtbj1-7	A0A0A6YVP1	T cell receptor beta joining 1-7 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.81	Trtbj1-4	A0A0A6YXY4	T cell receptor beta joining 1-4 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.81	Trtbj2-1	A0A0A6YWW0	T cell receptor beta joining 2-1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.82	Trtbj2-5	A0A0B4J1P2	T cell receptor beta joining 2-5 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.86	Trtbj1-6	A0A0A6YWT0	T cell receptor beta joining 1-6 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.87		A0ABJ3HP21	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.89	Traj43	A0A0G2JE21	T cell receptor alpha joining 43 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.90	Traj12	A0A0G2JDG3	T cell receptor alpha joining 12 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.91	Traj15	A0A0G2JG86	T cell receptor alpha joining 15 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.92	Ighj4	A0A0A6YXC7	Immunoglobulin heavy joining 4 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.93	Traj20	A0A0G2JFK9	T cell receptor alpha joining 20 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.94	Traj44	A0A0G2JE22	T cell receptor alpha joining 44 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.95	Traj37	A0A0G2JGK9	T cell receptor alpha joining 37 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.95	Traj13	A0A0G2JER2	T cell receptor alpha joining 13 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.96	Traj27	A0A0G2JFL6	T cell receptor alpha joining 27 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.97	Ighj1	A0A0A6YVQ4	Immunoglobulin heavy joining 1 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.97	Traj56	A0A0G2JDZ0	T cell receptor alpha joining 56 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.99	Traj21	A0A0G2JG57	T cell receptor alpha joining 21 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.99	Traj58	A0A0G2JFA3	T cell receptor alpha joining 58 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.99	Traj57	A0A0G2JH10	T cell receptor alpha joining 57 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
2.00	Gm66057	A0AAQ4VMP7	Predicted gene, 66057	0.0-2.0 kDa		standard full-blot and antibody-validation checks
2.00	Traj22	A0A0G2JDE9	T cell receptor alpha joining 22 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
2.00	Traj11	A0A0G2JE76	T cell receptor alpha joining 11 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
2.00	Traj50	A0A0G2JEE5	T cell receptor alpha joining 50 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
2.00	Traj9	A0A0G2JEW3	T cell receptor alpha joining 9 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.00	Traj25	A0A0G2JFP6	T cell receptor alpha joining 25 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.01	Traj24	A0A0G2JEV8	T cell receptor alpha joining 24 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.01	Traj31	A0A0G2JDF0	T cell receptor alpha joining 31 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.01	Traj41	A0A0G2JF06	T cell receptor alpha joining 41 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.02	Trgj2	A0A0G2JGW7	T cell receptor gamma joining 1 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.02	Traj34	A0A0G2JGM3	T cell receptor alpha joining 34 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.03	Traj33	A0A0G2JDY8	T cell receptor alpha joining 33 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.04	Traj4	A0A0G2JF02	T cell receptor alpha joining 4 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.04	Traj26	A0A0G2JGI7	T cell receptor alpha joining 26 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.04	Traj39	A0A0G2JDX5	T cell receptor alpha joining 39 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.05	Traj6	A0A0G2JEB4	T cell receptor alpha joining 6 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.06	Traj47	A0A0G2JDH7	T cell receptor alpha joining 47 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.07	Traj2	A0A0G2JDE4	T cell receptor alpha joining 2 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.07	Traj7	A0A0G2JFF4	T cell receptor alpha joining 7 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.08	Traj42	A0A0G2JFN0	T cell receptor alpha joining 42 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
2.08	Traj38	A0A0G2JFM0	T cell receptor alpha joining 38 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.10	Traj35	A0A0G2JFU1	T cell receptor alpha joining 35 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.10	Traj17	A0A0G2JEC1	T cell receptor alpha joining 17 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.11		P83891	Implantin (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.11	Traj23	A0A0G2JDS9	T cell receptor alpha joining 23 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.11	Traj30	A0A0G2JDP5	T cell receptor alpha joining 30 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.11	Traj16	A0A0G2JEK3	T cell receptor alpha joining 16 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.12	Traj29	A0A0G2JFV0	T cell receptor alpha joining 29 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.12	Traj48	A0A0G2JG88	T cell receptor alpha joining 48 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.13	Trgj4	A0A0G2JF87	T cell receptor gamma joining 4 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.15		A0ABJ3HKJ1	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.15	Traj5	A0A0G2JFC2	T cell receptor alpha joining 5 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.16	Traj53	A0A0G2JGM9	T cell receptor alpha joining 53 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.16	Traj45	A0A0G2JFC4	T cell receptor alpha joining 45 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.16	Traj46	A0A0G2JGR3	T cell receptor alpha joining 46 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.17	Traj32	A0A0G2JF50	T cell receptor alpha joining 32 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.17	Traj40	A0A0G2JH15	T cell receptor alpha joining 40 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.17	Trgj3	A0A0G2JEl4	T cell receptor gamma joining 3 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.18	Traj18	A0A0G2JG46	T cell receptor alpha joining 18 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.19	Traj52	A0A0G2JFS6	T cell receptor alpha joining 52 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.19	Traj28	A0A0G2JFW3	T cell receptor alpha joining 28 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.19	Traj49	A0A0G2JFY2	T cell receptor alpha joining 49 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.24	Traj59	A0A0G2JF27	T cell receptor alpha joining 59 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.26	Trdj2	A0A0G2JFW2	T cell receptor delta joining 2 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.28		P83855	Putative sperm adenylyate cyclase (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.29	Traj19	A0A0G2JD6	T cell receptor alpha joining 19 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.32	Gm69672	A0A9L6KDT7	Predicted gene, 69672	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.33	Mcp13	P21843	Mast cell protease 3 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.44	Gm68923	A0A9L6KED7	Predicted gene, 68923	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.45	Traj3	A0A0G2JFD7	T cell receptor alpha joining 3 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.53	Gm14180	B1AQAO	Predicted gene 14180	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.55	Gm68747	A0AAQ4VMP9	Predicted gene, 68747	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.57	Gm74510	A0AAQ4VMV8	Predicted gene, 74510	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.68		A0ABJ3HP41	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.76	Gm64583	A0AA9WWD5	Predicted gene, 64583	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.89	Gm59508	A0A9L6KED2	Predicted gene, 59508	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.91	Gm68765	A0A9L6KDW9	Predicted gene, 68765	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.92		A0ABA7IXB2	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.94	Gm58353	A0AAQ4VMT8	Predicted gene, 58353	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.01		A0ABJ3HP54	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.05		A0ABJ3HNV8	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.14		A0ABJ3HKL9	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.14	Gm62004	A0AAA9WUT7	Predicted gene, 62004	2.0-4.0 kDa		standard full-blot and antibody-validation checks

Reported western-blot malpractice cases mapped to MW bins

These are reported public cases supplied in the malpractice input TSV. They are evidence examples for integrity-audit results sections, not substitution recommendations.

Case	Claim/protein	MW bin	Reported issue	Results sentence
ORI_Eckert_JBC2014_MEK3	MEK3 western blot	38.0-40.0 kDa	ORI reported MEK3 bands were compiled from a source image representing p44 expression in an unrelated experiment.	A public ORI case documents a human-cell western blot malpractice example in the 38-40 kDa bin, where MEK3-labeled bands were compiled from an unrelated p44 source image.
ORI_Eckert_JBC2014_p38d	p38δ western blot	40.0-42.0 kDa	ORI reported p38δ bands were compiled from a source image representing β-actin expression in an unrelated experiment.	The 40-42 kDa bin is supported by a public ORI case in which p38δ-labeled bands were compiled from a β-actin source image.
ORI_Eckert_PLoS2012_TAM67	TAM67-FLAG western blot	48.0-50.0 kDa	ORI reported TAM67-FLAG bands were generated using unrelated bands from a source image representing Cyclin A expression.	A public ORI case maps a relabeled western blot problem to the 48-50 kDa neighborhood, involving Cyclin A source material used for a TAM67-FLAG claim.
ORI_Yang_GAPDH_NFYA	GAPDH loading-control bands	36.0-38.0 kDa	ORI reported claimed GAPDH loading-control bands were actually a prominent nonspecific/background band from an NF-YA western blot.	The 36-38 kDa neighborhood is supported by an ORI-described case where a nonspecific NF-YA blot band was claimed as GAPDH.
CancerGeneTherapy_Liu2021_CTBP2_GAPDH	CTBP2 and GAPDH western blots	48.0-50.0 kDa	Retraction note reported apparent similarity in western blots presented for different proteins, including CTBP2 and GAPDH.	A retraction notice maps an apparent different-protein western blot similarity to the CTBP2/GAPDH region, spanning roughly the 36-38 and 48-50 kDa neighborhoods.
CancerGeneTherapy_Liu2021_CyclinD1_CDK4	Cyclin D1 and CDK4 western blots	34.0-36.0 kDa	Retraction note reported apparent similarity in western blots presented for Cyclin D1 and CDK4.	A retraction notice maps an apparent different-protein western blot similarity to the 32-36 kDa cell-cycle neighborhood involving Cyclin D1 and CDK4.
Das_UConn_Garlic_AKT	Akt / p-Akt western blots in Figure 6	54.0-56.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 54-56 kDa signaling region is supported as an audit-priority bin by the garlic-paper retraction, where Akt/p-Akt figure panels were within figures reported to contain fabricated data.
Das_UConn_Garlic_FOXO1	FoxO1 / p-FoxO1 western blots in Figure 6	70.0-72.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 70 kDa transcription-factor signaling region can be referenced as an audit-priority bin because FoxO1 panels occurred in a figure reported to contain fabricated data.
Das_UConn_Garlic_NFKB_PPAR_Glut4	Nrf2, NF-κB p65, PPARα/δ, Glut-4 western blots in Figures 7 and 8	52.0-54.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 50-56 kDa metabolic/signaling region is an audit-priority region in the garlic-paper case because PPAR and Glut-4 panels were in figures reported to contain fabricated data.
Ilangovan_HSP25_HSP27	Hsp25/Hsp27-family western blots	22.0-24.0 kDa	Public investigation materials described Hsp-family western blot relabeling and unreliable/falsified blot images.	The 22-24 kDa small heat-shock-protein neighborhood can be cited as an audit-risk bin based on public investigation materials describing Hsp-family western blot relabeling.

Integrity-audit checklist

- Ask for the full, uncropped blot with ladder and lane labels. Cropped panels are not enough.
- Confirm expected molecular weight and apparent molecular weight. Do not treat molecular-weight agreement as proof of identity.
- For configured marker proteins, request source-imager metadata and original antibody-validation records.
- Check whether a marker band is near common markers such as GAPDH, actin, tubulin, HSPs, lamins, VDAC, PCNA, or other configured controls.
- For signaling proteins, verify phospho/total controls, controls for activation state, and an independent identity control when central to the claim.
- In crowded or high-audit bins, prioritize raw-data provenance, full blot context, and orthogonal validation.
- Interpret mapped malpractice cases only as documented audit precedents. They do not imply that proteins in the same bin are interchangeable.

Generated files

- proteome_wb_integrity_atlas.csv - complete per-protein atlas sorted by theoretical MW.
- mw_bin_summary.csv - complete molecular-weight-bin audit summary.
- reported_malpractice_bins.csv - generated when a malpractice TSV is supplied.
- plots/ - four standalone PNG figures embedded in this report.
- western_blot_integrity_atlas_full_report_with_plots.pdf - this consolidated report.
- western_blot_integrity_atlas_report.pdf - byte-identical compatibility copy of the consolidated report.

Input configuration

Input	File
Proteome FASTA	mouse_UP000000589_10090.fasta
Marker file	mouse_marker_genes.tsv
Signaling-prefix file	mouse_signaling_prefixes.tsv
Family-pattern file	mouse_family_patterns.tsv
Species/report label	Mus musculus